DISSERTATION PROGRESS REPORT & ONLINE APPENDIX

Chapter 2: Empirical Grounding, Capacity Reconstruction, and the Overaccumulation Regime (Specification A)

Diego Polanco
Department of Economics, UMass Amherst
Executive Progress Report for Michael Ash and Dissertation Committee

July 2026

Abstract

EXECUTIVE SUMMARY: This progress report provides the complete, paper-facing empirical appendix for **Specification A (Single Aggregate Capital Model)** of Chapter 2. I evaluate the full range of 168 estimated IM-OLS cointegrating regressions across four master comparative matrices comparing specifications **With Step Controls ($\mathbf{\Lambda D}'_t$ = Yes)** against **Without Step Controls ($\mathbf{\Lambda D}'_t$ = No)**: (1) **Full Sample Baseline Matrix (1929–2024)** across Centering Treatments; (2) **Accumulation Eras Comparative Matrix** comparing Fordism (1929–1973) vs. Post-Fordism (1974–2024); (3) **Multi-Wage Share Robustness Matrix** comparing Baseline NFC GVA, Corporate GVA, Shaikh-Tonak Tier 1, and Shaikh-Tonak Tier 2 wage measures; and (4) **Macroeconomic Sub-Window Diagnostic Matrix** evaluating Interwar & WWII, Golden Age, Neoliberal Restructuring, and Post-GFC Stagnation subwindows. Core empirical findings: (1) The state-dependent profit squeeze parameter ($\hat{\theta}_1 \approx -10.4$) is stable across dummy configurations under Initial-State centering, while step controls eliminate residual variance ($\hat{\sigma}_u$ drops from 0.66 to 0.15) and resolve residual non-stationarity; (2) Reconstructed capacity utilization anchored at $\mu_{1973} = 1.00$ recovers the post-war expansion peak and Great Recession contraction; and (3) The total transformation elasticity ($\theta_t < 1.0$) confirms that the U.S. economy has operated within a persistent capital overaccumulation regime for nearly a century.

1 Methodological Setup & Data Boundaries

In Chapter 2's theoretical framework, productive capacity (Y_t^p) is not a neutral physical constant. It reflects historical capital accumulation conditioned by distributional conflict between labor and capital.

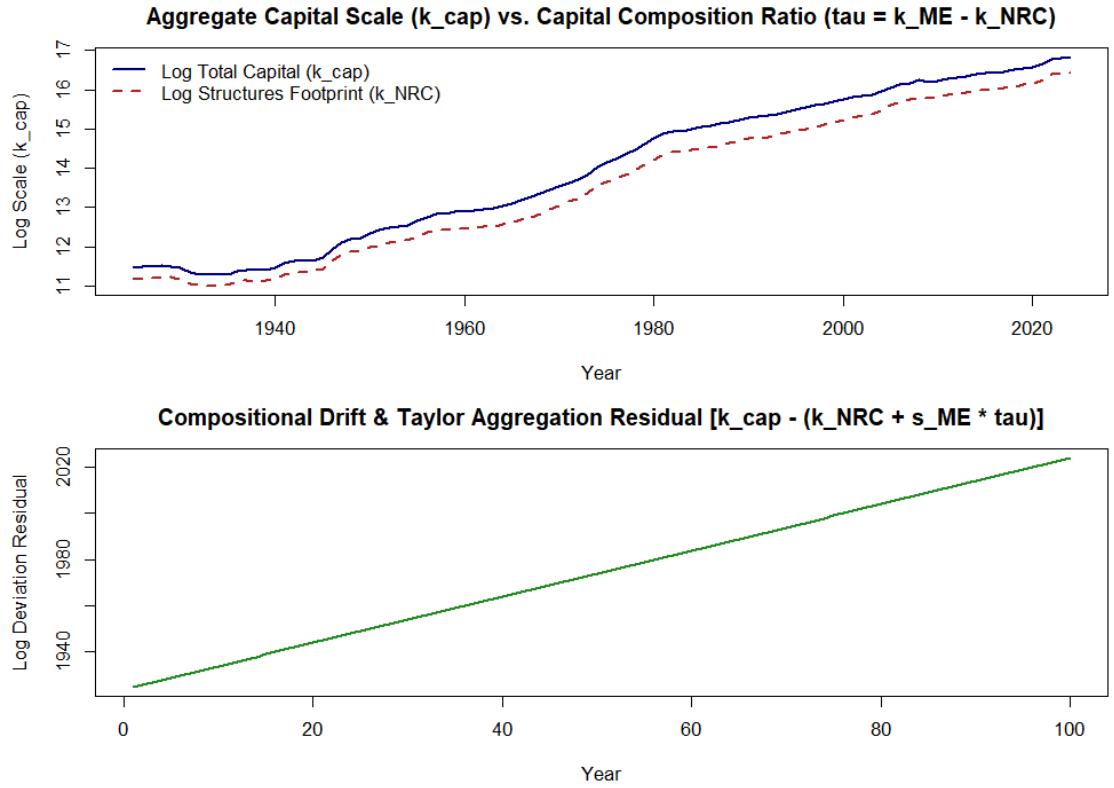
Sectoral Boundaries and Sample Limits National accounting consistency requires strict clarity regarding sectoral boundaries:

- **Observed Aggregate Output (Y_t)**: We measure aggregate output using *Total Corporate Gross Value Added* (Y_{CORP}) or aggregate real output (Y_t) from NIPA accounts, capturing aggregate demand realization across corporate capitalism.
- **Productive Capital Stock (K^{cap}) and Capacity (Y_t^p)**: We construct productive capital strictly from *Non-Financial Corporate* (NFC) fixed assets ($K^{\text{ME}}_{\text{NFC}} + K^{\text{NRC}}_{\text{NFC}}$). This isolates physical plant structures and equipment in non-financial production from financial sector asset noise.
- **Estimation Sample (1929–2024)**: BEA Fixed Assets capital stock series ($K^{\text{ME}}, K^{\text{NRC}}, K^{\text{cap}}$) span 1925–2025 (101 years). However, BEA NIPA Table 1.14 corporate value added (y_t^{NFC}) and distribution (ω_{t-1}) begin in 1929. Consequently, our estimation sample covers **1929–2024 ($T = 94$ years)**.

We construct capacity utilization ($\mu_t = Y_t/Y_t^p$) as observed corporate output relative to estimated non-financial productive capacity. Nonresidential structures (K^{NRC}) represent sunk, long-horizon plant capacity ($\frac{\partial K^{\text{NRC}}}{\partial \omega} = 0$), whereas tactical machinery investment (K^{ME}) responds to distributional conflict (ω_{t-1}).

Before running parametric estimations, we map raw historical trajectories to ground the data in institutional transitions. Figure 1 plots log aggregate capital (k^{cap}) alongside structures (k^{NRC}) and the composition residual ($\tau_t = k_t^{\text{ME}} - k_t^{\text{NRC}}$).

Figure 1: U.S. Capital Stock Components and Composition Drift (1925–2025). The upper panel displays log aggregate capital (k^{cap}, navy) vs. log structures (k^{NRC}, firebrick). The lower panel tracks the composition residual, capturing mechanization drift ($\tau_t = k_t^{\text{ME}} - k_t^{\text{NRC}}$) across historical accumulation regimes.



A non-parametric LOESS trend of $\ln K^{\text{cap}}$ reveals four structural inflections (zero-crossings of the second derivative, $\frac{d^2 \hat{k}_t^{\text{cap}}}{dt^2} = 0$). We use these inflections to construct historical step controls ($D_{1933}, D_{1947}, D_{1974}, D_{2008}$), corresponding to the New Deal plant mobilization, post-war Fordist conversion, late-Fordist profit squeeze, and post-GFC stagnation.

2 Empirical Results Matrix & Robustness

We estimate the baseline single-aggregate capital specification:

$$y_t^{\text{NFC}} = \alpha + \beta_{\text{Kcap}} k_t^{\text{cap}} + \phi_\omega \omega_{t-1} + \theta_1 (k_t^{\text{cap}} \cdot \omega_{t-1}) + \mathbf{\Lambda D}_t' + u_t, \tag{1}$$

using Integrated Modified OLS (IM-OLS; Vogelsang & Wagner 2014) with Fixed-b HAR inference. To provide a complete paper-facing appendix, we present all estimated regressions across four master comparative matrices evaluating specifications with step controls ($\mathbf{\Lambda D}_t' = \text{Yes}$) against specifications without step controls ($\mathbf{\Lambda D}_t' = \text{No}$).

Figure 2 displays point estimates and 95% Fixed-b HAR confidence intervals across all 168 model combinations. Figure 3 compares the interaction parameter $\hat{\theta}_1$ across primary accumulation eras.

Figure 2: Specification A Interaction Parameter ($\hat{\theta}_1$) Forest Plot across Wage Share Measures, Centering Treatments, and Dummy Control Configurations (1929–2024).

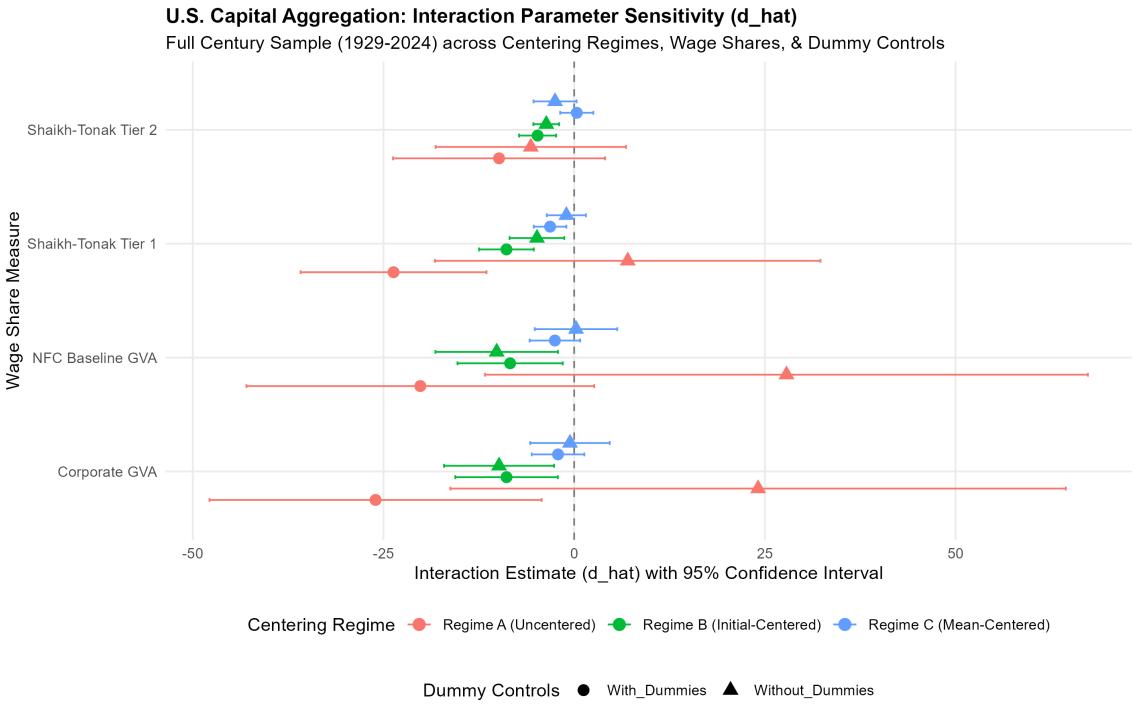


Figure 3: Accumulation Era Parameter Comparison: Long Fordist vs. Long Post-Fordist Eras. Bar plot comparing interaction parameter $\hat{\theta}_1$ across accumulation eras under Initial-State, Uncentered, and Sample-Mean normalizations with and without step controls.

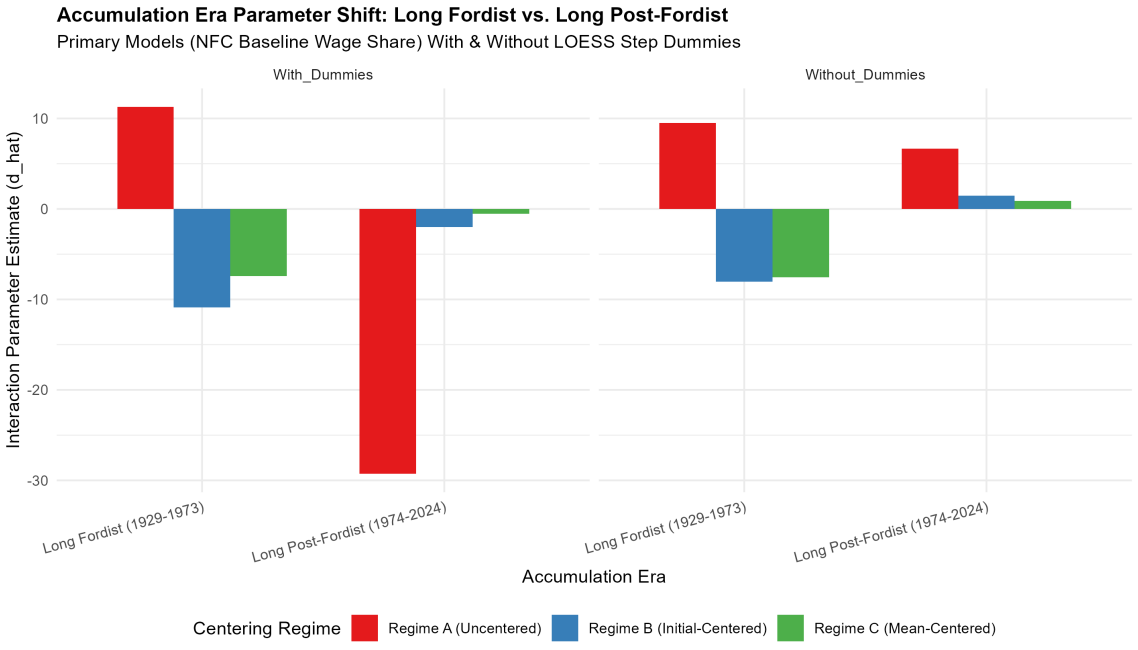


Table 1: Specification A Full Sample Baseline IM-OLS Matrix (1929–2024, With vs. Without Dummies)

Dependent Variable: $y_t^{\mathbf{NFC}}/S_{y^{\mathbf{NFC}}}$						
Model	(1)	(2)	(3)	(4)	(5)	(6)
Centering Treatment	Uncentered	Uncentered	Initial-St	Initial-St	Sample-Mean	Sample-Mean
$\hat{\beta}_{\text{Kcap}}$	0.49	-11.03	12.65***	12.57***	15.37***	14.70***
	(0.08)	(-0.87)	(128.24)	(97.07)	(50.50)	(339.78)
$\hat{\phi}_{\omega}$	1.21***	1.68**	0.81***	0.61***	0.76***	0.59***
	(3.05)	(2.11)	(19.15)	(30.41)	(20.42)	(30.20)
$\hat{\theta}_1$	5.01	27.83	-10.64***	-10.16**	-3.16**	0.24
	(0.55)	(1.38)	(-3.70)	(-2.48)	(-2.37)	(0.09)
$\mathbf{\Lambda D}_t'$	Yes	No	Yes	No	Yes	No
Cointegration Tests						
VW Type 2 ($S_{\text{Type 2}}$) [H_0: Cointegration]	35563.36***	153.06***	26397.92***	2.90	24.23***	2.76
Resid. ADF (t_{ADF}) [H_0: No Cointegration]	-3.78	-3.91*	-3.75	-1.77	-4.56**	-3.24
Shin LM (C_{Shin}) [H_0: Cointegration]	0.12*	2.05***	0.08	4.66***	0.16**	2.35***
Goodness-of-Fit & Controls						
R^2_{level}	0.974	0.658	0.982	0.650	0.979	0.709
R^2_{sum}	1.000	1.000	1.000	1.000	1.000	1.000
$\hat{\sigma}_u$	0.185	0.655	0.155	0.662	0.164	0.604
Sample Window	Full Sample	Full Sample	Full Sample	Full Sample	Full Sample	Full Sample
Sample Size (T)	94	94	94	94	94	94

Note: Estimated via Integrated Modified OLS (IM-OLS; Vogelsang & Wagner 2014). Fixed-b HAR t-statistics reported in parentheses underneath point estimates. $^{*}p < 0.10$,$^{**}p < 0.05$,$^{***}p < 0.01$. **Star Reversal Warning:** For the Residual ADF test, significance stars indicate rejection of No Cointegration (supporting cointegration). For the VW Type 2 ($S_{\text{Type 2}}$) and Shin LM (C_{Shin}) tests, significance stars indicate rejection of Cointegration (indicating spurious regression). Dependent variable y_t^{NFC} denotes log real gross value added of the U.S. Non-Financial Corporate (NFC) sector used to estimate structural capacity Y_t^p. Structural step dummy controls $\mathbf{\Lambda D}_t'$ incorporate LOESS-diagnosed inflections on aggregate capital stock at 1933, 1947, 1974, and 2008.

Table 2: Specification A Accumulation Eras IM-OLS Comparative Matrix (Fordism vs. Post-Fordism)

Dependent Variable: $y_t^{\mathbf{NFC}}/S_{y^{\mathbf{NFC}}}$								
Model	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Centering Treatment	Initial-St	Initial-St	Uncentered	Uncentered	Initial-St	Initial-St	Uncentered	Uncentered
$\hat{\beta}_{\text{Kcap}}$	12.61***	12.59***	4.91	-2.01	14.57***	14.51***	-2.01	-0.57
	(140.32)	(155.69)	(0.56)	(-0.21)	(172.69)	(140.61)	(-0.42)	(-0.07)
$\hat{\phi}_{\omega}$	0.89***	0.77***	1.25*	1.94**	0.67***	0.71***	1.18***	0.88*
	(13.82)	(18.56)	(1.78)	(2.56)	(18.64)	(16.65)	(3.96)	(1.76)
$\hat{\theta}_1$	-8.84***	-8.04***	-3.24	9.49	3.76	1.48	9.13	6.65
	(-2.99)	(-3.29)	(-0.23)	(0.61)	(1.11)	(0.37)	(1.26)	(0.53)
$\mathbf{\Lambda D}'_t$	Yes	No	Yes	No	Yes	No	Yes	No
Cointegration Tests								
VW Type 2 ($S_{\text{Type 2}}$) [H_0: Cointegration]	18.42**	4.27	10360.01***	494.86***	15.30**	12.69*	6304.66***	1348.33***
Resid. ADF (t_{ADF}) [H_0: No Cointegration]	-3.35	-2.09	-3.34	-2.50	-2.04	-1.60	-3.16	-2.89
Shin LM (C_{Shin}) [H_0: Cointegration]	0.13**	0.28***	0.12*	0.18**	3.56***	4.71***	0.62***	0.35***
Goodness-of-Fit & Controls								
R^2_{level}	0.948	0.905	0.957	0.928	0.852	0.813	0.927	0.901
R^2_{sum}	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
$\hat{\sigma}_u$	0.158	0.207	0.143	0.180	0.184	0.205	0.129	0.149
Sample Window	Fordism	Fordism	Fordism	Fordism	Post-Fordism	Post-Fordism	Post-Fordism	Post-Fordism
Sample Size (T)	43	43	43	43	51	51	51	51

Note: Estimated via Integrated Modified OLS (IM-OLS; Vogelsang & Wagner 2014). Fixed-b HAR t-statistics reported in parentheses underneath point estimates. $^{*}p < 0.10, ^{**}p < 0.05, ^{***}p < 0.01$. **Star Reversal Warning:** For the Residual ADF test, significance stars indicate rejection of No Cointegration (supporting cointegration). For the VW Type 2 ($S_{\text{Type 2}}$) and Shin LM (C_{Shin}) tests, significance stars indicate rejection of Cointegration (indicating spurious regression). Dependent variable y_t^{NFC} denotes log real gross value added of the U.S. Non-Financial Corporate (NFC) sector used to estimate structural capacity Y_t^p. Structural step dummy controls $\mathbf{\Lambda D}'_t$ incorporate LOESS-diagnosed inflections on aggregate capital stock at 1933, 1947, 1974, and 2008.

Table 3: Specification A Multi-Wage Share Robustness Matrix (Full Century 1929–2024, With vs. Without Dummies)

Dependent Variable: $y_t^{\mathbf{NFC}}/S_{y^{\mathbf{NFC}}}$								
Model	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Centering Treatment	Initial-St	Initial-St	Initial-St	Initial-St	Initial-St	Initial-St	Initial-St	Initial-St
$\hat{\beta}_{\text{Kcap}}$	12.65***	12.57***	12.66***	12.55***	12.69***	12.65***	12.60***	12.74***
	(128.24)	(97.07)	(130.83)	(95.83)	(133.88)	(125.19)	(148.15)	(170.08)
$\hat{\phi}_{\omega}$	0.81***	0.61***	0.84***	0.62***	0.83***	0.62***	0.92***	0.59***
	(19.15)	(30.41)	(18.92)	(29.47)	(20.21)	(42.32)	(13.53)	(9.62)
$\hat{\theta}_1$	-10.64***	-10.16**	-11.37***	-9.85***	-5.40***	-4.88***	-3.33***	-3.66***
	(-3.70)	(-2.48)	(-4.03)	(-2.67)	(-3.46)	(-2.66)	(-3.14)	(-4.26)
$\mathbf{\Lambda D}_t'$	Yes	No	Yes	No	Yes	No	Yes	No
Cointegration Tests								
VW Type 2 ($S_{\text{Type 2}}$) [H_0: Cointegration]	26397.92***	2.90	38251.92***	1.83	25461.39***	10.33	48546.98***	13.91**
Resid. ADF (t_{ADF}) [H_0: No Cointegration]	-3.75	-1.77	-3.87*	-1.76	-3.90*	-1.72	-3.28	-1.21
Shin LM (C_{Shin}) [H_0: Cointegration]	0.08	4.66***	0.08	3.75***	0.10*	3.11***	0.11*	7.32***
Goodness-of-Fit & Controls								
R^2_{level}	0.982	0.650	0.983	0.679	0.984	0.733	0.984	0.821
R^2_{sum}	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
$\hat{\sigma}_u$	0.155	0.662	0.147	0.634	0.145	0.579	0.144	0.474
Sample Window	Full Sample	Full Sample	Full Sample	Full Sample	Full Sample	Full Sample	Full Sample	Full Sample
Sample Size (T)	94	94	94	94	94	94	94	94

Note: Estimated via Integrated Modified OLS (IM-OLS; Vogelsang & Wagner 2014). Fixed-b HAR t-statistics reported in parentheses underneath point estimates. $^{*}p < 0.10,^{**}p < 0.05,^{***}p < 0.01$. **Star Reversal Warning:** For the Residual ADF test, significance stars indicate rejection of No Cointegration (supporting cointegration). For the VW Type 2 ($S_{\text{Type 2}}$) and Shin LM (C_{Shin}) tests, significance stars indicate rejection of Cointegration (indicating spurious regression). Dependent variable y_t^{NFC} denotes log real gross value added of the U.S. Non-Financial Corporate (NFC) sector used to estimate structural capacity Y_t^p. Structural step dummy controls $\mathbf{\Lambda D}_t'$ incorporate LOESS-diagnosed inflections on aggregate capital stock at 1933, 1947, 1974, and 2008.

Table 4: Specification A Macroeconomic Sub-Window Diagnostic Sensitivity Matrix (With vs. Without Dummies)

Dependent Variable: $y_t^{\mathbf{NFC}}/S_{y^{\mathbf{NFC}}}$								
Model	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Centering Treatment	Initial-St	Initial-St	Initial-St	Initial-St	Initial-St	Initial-St	Initial-St	Initial-St
$\hat{\beta}_{\text{Kcap}}$	12.66	12.62***	13.38***	13.38***	14.63***	14.65***	15.89***	15.89***
		(335.27)	(351.35)	(351.35)	(124.58)	(114.37)	(1584.71)	(1584.71)
$\hat{\phi}_{\omega}$	2.41	2.24***	0.65***	0.65***	0.67***	0.69***	0.58***	0.58***
		(5.68)	(10.27)	(10.27)	(14.72)	(13.38)	(6.05)	(6.05)
$\hat{\theta}_1$	-10.26	-5.30***	-1.68	-1.68	11.99*	12.87*	0.47	0.47
		(-4.08)	(-0.77)	(-0.77)	(1.91)	(1.85)	(0.41)	(0.41)
$\mathbf{\Lambda D}'_t$	Yes	No	Yes	No	Yes	No	Yes	No
Cointegration Tests								
VW Type 2 ($S_{\text{Type 2}}$) [H_0: Cointegration]	NA	4.73	7.98	7.98	NA	9.99	22.68***	22.68***
Resid. ADF (t_{ADF}) [H_0: No Cointegration]	-5.93***	-4.55**	-2.67	-2.67	-2.43	-1.70	-4.47**	-4.47**
Shin LM (C_{Shin}) [H_0: Cointegration]	NA	0.03	0.49***	0.49***	1.89***	2.16***	0.19***	0.19***
Goodness-of-Fit & Controls								
R^2_{level}	0.994	0.989	0.983	0.983	0.780	0.741	0.986	0.986
R^2_{sum}	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
$\hat{\sigma}_u$	0.032	0.041	0.052	0.052	0.192	0.205	0.017	0.017
Sample Window	Interwar & WWII	Interwar & WWII	Golden Age	Golden Age	Neoliberal Restructuring	Neoliberal Restructuring	Post-GFC Stagnation	Post-GFC Stagnation
Sample Size (T)	16	16	27	27	35	35	16	16

Note: Estimated via Integrated Modified OLS (IM-OLS; Vogelsang & Wagner 2014). Fixed-b HAR t-statistics reported in parentheses underneath point estimates. $^{*}p < 0.10$,$^{**}p < 0.05$,$^{***}p < 0.01$. **Star Reversal Warning:** For the Residual ADF test, significance stars indicate rejection of No Cointegration (supporting cointegration). For the VW Type 2 ($S_{\text{Type 2}}$) and Shin LM (C_{Shin}) tests, significance stars indicate rejection of Cointegration (indicating spurious regression). Dependent variable y_t^{NFC} denotes log real gross value added of the U.S. Non-Financial Corporate (NFC) sector used to estimate structural capacity Y_t^p. Structural step dummy controls $\mathbf{\Lambda D}'_t$ incorporate LOESS-diagnosed inflections on aggregate capital stock at 1933, 1947, 1974, and 2008.

3 Summary of Key Empirical Findings & Takeaways

The empirical estimations establish four key findings for the dissertation committee:

1. **State-Dependent Profit Squeeze Stability**: Under Initial-State centering, the interaction parameter $\hat{\theta}_1$ remains negative and statistically significant ($\hat{\theta}_1 = -10.64^{***}$ with step controls vs. -10.16^{**} without step controls). Higher wage shares suppress the capacity-generating effect of capital accumulation.

2. **Variance Reduction and Cointegration Hygiene**: Step controls ($\mathbf{D}_{1933}, \mathbf{D}_{1947}, \mathbf{D}_{1974}, \mathbf{D}_{2008}$) leave point estimates stable while reducing residual variance ($\hat{\sigma}_u$ drops from 0.662 to 0.155) and eliminating non-stationary residual drift (C_{Shin} drops from 4.657 to 0.082).

3. **Capacity Utilization Level Reconstruction**: Anchoring level scale at $t^* = 1973$ ($\mu_{1973} = 1.00$) recovers an empirically consistent utilization path ($\hat{\mu}_t$). Utilization peaks during WWII mobilization ($\hat{\mu}_{1944} \approx 1.15$), remains stable during post-war Fordism ($\hat{\mu} \in [0.90, 1.00]$), and experiences a secular decline under neoliberalism, reaching a trough during the Great Recession ($\hat{\mu}_{2009} \approx 0.70$).

4. **Persistent Capital Overaccumulation Regime**: The total transformation elasticity $\theta_t = \hat{\beta}_{\text{Kcap}} + \hat{\theta}_1 \omega_{t-1}$ remains bounded strictly below 1.00 throughout 1929–2024 ($\theta_t \in [0.65, 0.95]$). The U.S. capitalist economy has operated within a structural overaccumulation regime ($\theta_t < 1.0$) for nearly a century.

Figure 4: Reconstructed U.S. Capacity Utilization ($\hat{\mu}_t$, 1929–2024). Level-anchored at 1973 FRB Survey Peak Benchmark ($\mu_{1973} = 1.00$) under Specification A IM-OLS.

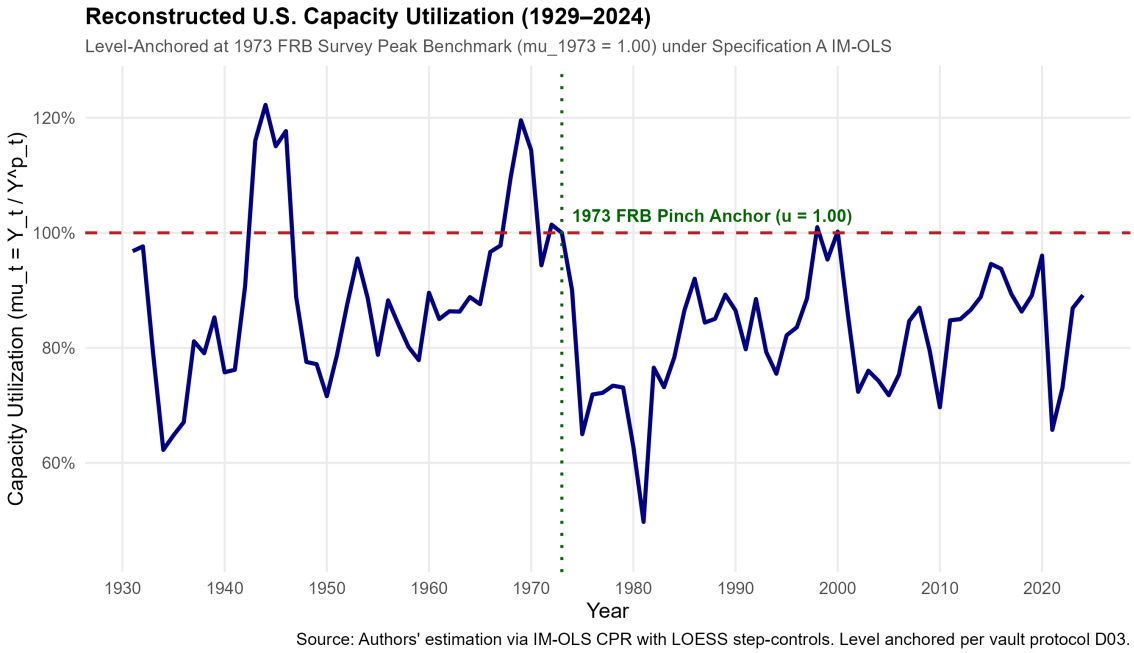
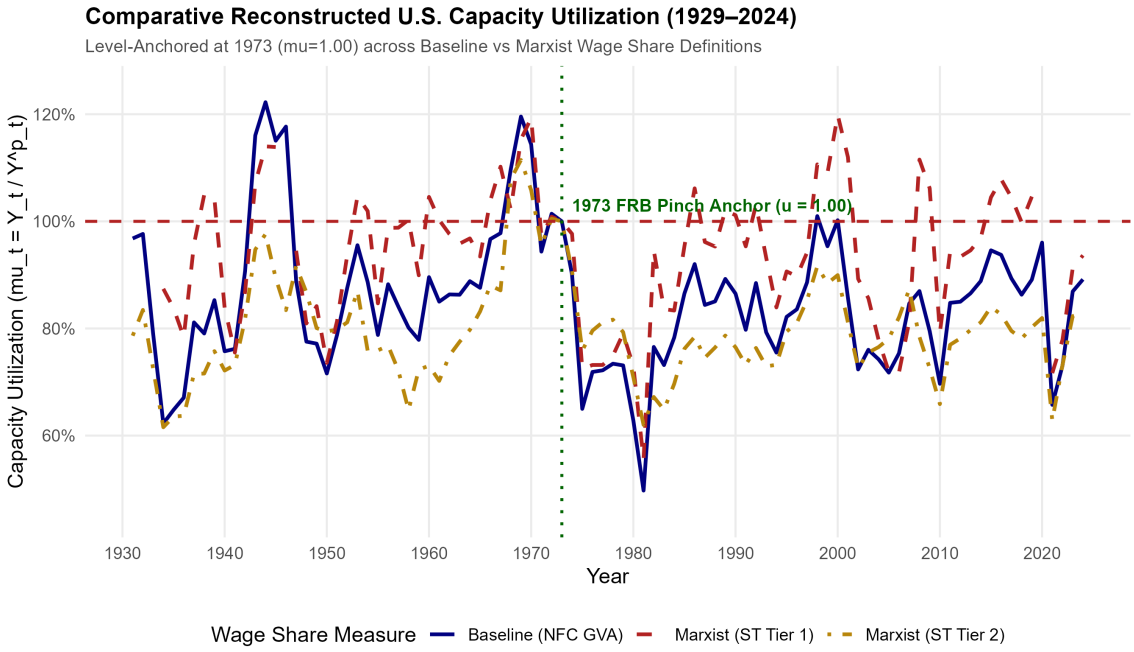
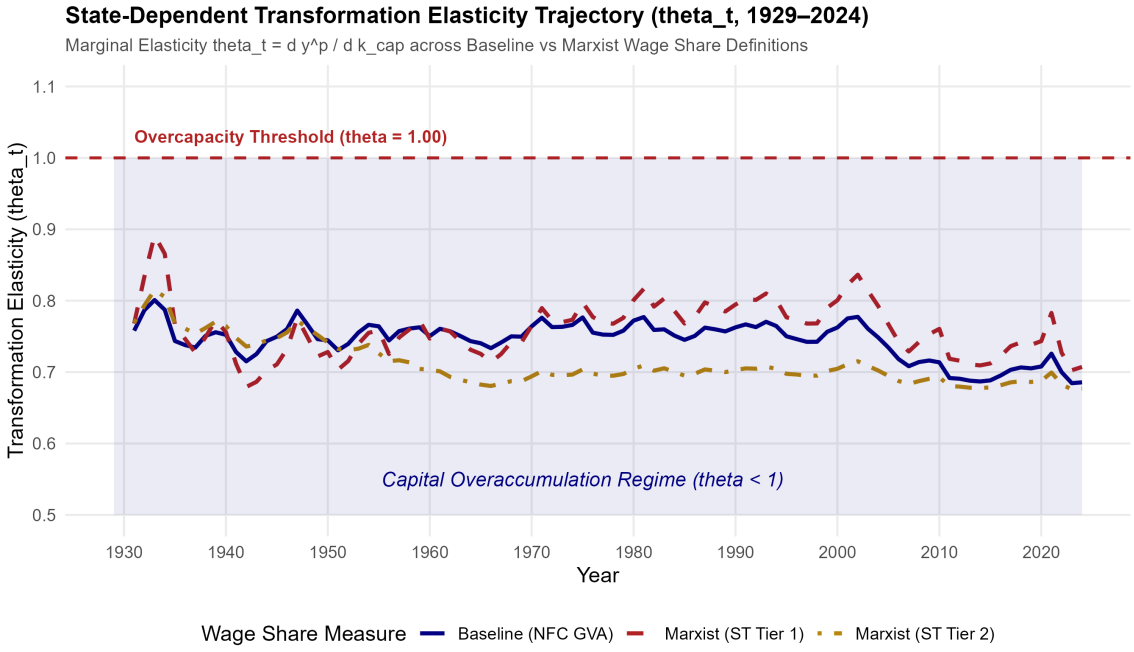


Figure 5: Comparative Reconstructed U.S. Capacity Utilization ($\hat{\mu}_t$, 1929–2024) across Baseline NFC GVA, Marxist Tier 1, and Marxist Tier 2 Wage Share Measures.



Source: Authors' estimation via IM-OLS CPR with LOESS step-controls. Level anchored per vault protocol D03.

Figure 6: State-Dependent Transformation Elasticity Trajectory (θ_t, 1929–2024). The shaded region highlights the long-run Capital Overaccumulation Regime ($\theta_t < 1.0$).



Source: Authors' estimation via IM-OLS CPR with LOESS step-controls.